

Ichthyoplankton sampling design to monitor marine fish populations and communities

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ABSTRACT

The ability to assess, model, predict and manage the impacts of climate change and other anthropogenic stressors on marine ecosystems depends on having adequate ecological time series. Unfortunately the development of ecological time series considerably lags those for the physics and chemistry of the oceans. Ichthyoplankton time series are proposed here to fill this gap in ocean observations. Marine fish species spanning a wide range of families, habitats, feeding guilds, and trophic levels broadcast large numbers of their reproductive products into the open waters. For a limited period, the larvae generally reside in the upper 200 m of the water column, where they may be quantitatively sampled with plankton nets. Larval abundance provides a relative index for adult spawning stock biomass, enabling diverse fish communities to be monitored quantitatively by relatively simple means. Recent analyses of the ichthyoplankton time series extending back to 1951 from the California Cooperative Oceanic Fisheries Investigations (CalCOFI) program indicate that non-commercial as well as commercially-exploited taxa have experienced dramatic change in recent decades. The CalCOFI data set is re-sampled here to show that a reduced sampling program—one based on a few stations along a single transect (cf. > 50 stations along 6 transects for CalCOFI) or one based on a shorter time series—can, within limits, obtain similar single-species and multivariate patterns of abundance. Ichthyoplankton survey programs may thus provide the basis for a global system of ocean ecological observations in addition to their primary use today for fisheries stock assessment.

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1. Introduction

Revelle and Seuss [37] famously noted that “human beings are now carrying out a large-scale geophysical experiment of a kind that could not have happened in the past nor be reproduced in the future” and pointed to the need to “adequately document” it. More than half a century later, the increase in global atmospheric CO₂ concentrations has indeed been well documented, and satellites, profiling Argo floats and other sensor arrays provide global coverage of changing physical and chemical conditions in the world's oceans. However, although the Global Ocean Observing System (GOOS) was established in the 1990s, time series for the potentially changing ecology of the global ocean remain a patchwork [15,16].

Marine fishes would seem a prime candidate for inclusion in

ocean observation programs, being ubiquitous in the world's oceans, diverse, ecologically critical, and highly valued economically, socially and culturally. In fact, however, they are poorly represented in most ocean observing systems [15,16]. There is a global system of reporting marine fishery landings as well as many fishery assessment surveys, but commercial fisheries comprise a small sub-set of the approximately 28,000 marine fish species [33]. Landings statistics are fraught with biases due to the influence of market conditions, changing technology, and regulations. Comparison of landings data with fishery-independent time series has shown that commercial time series can be highly misleading as indices of population trends [17].

A vast literature, global in extent and extending back for decades, documents the response of commercial fish populations to climate [8,14]. Likewise, there is growing evidence that non-commercial fish populations may vary substantially due to the impacts of climate [11,20,24,35], deoxygenation [19], pollution [9] and other anthropogenic stressors. Fishes appear to be sensitive indicators of ecosystem change, but without sustained ocean time series, changes to fish communities will likely not be detected, and

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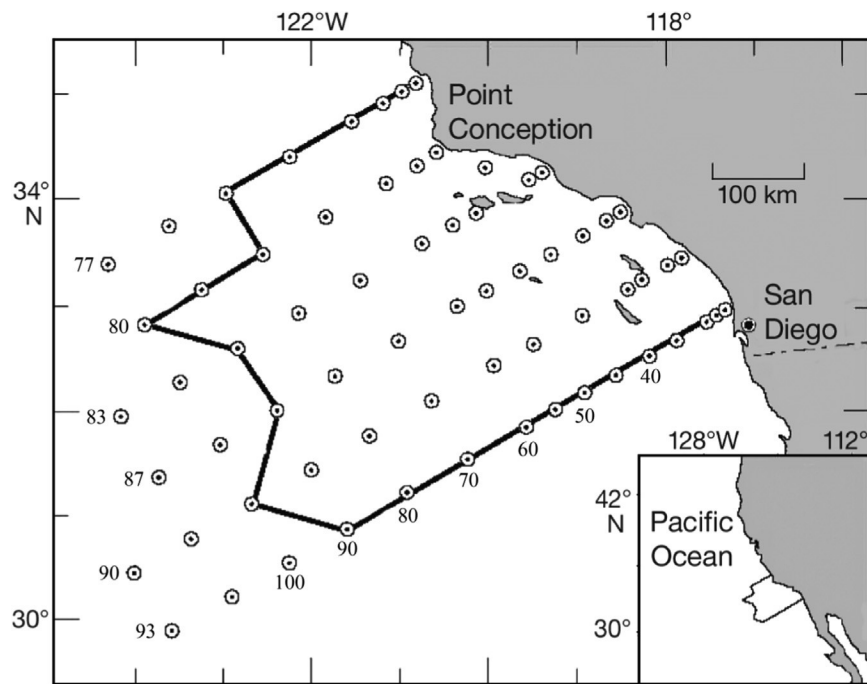


Fig. 1. The CalCOFI core survey design showing line (or transect) and station numbers. The 51 stations within the dark lines were most consistently sampled and used in analyses. Sub-sampling was carried out to stations 40, 60, or to the end of each transect.

Table 1

The common and scientific names and habitat of the 12 most abundant taxa in the CalCOFI ichthyoplankton data set used in time series analyses.

Common name	Scientific name	Habitat
Sand dabs	<i>Citharichthys</i> spp.	Demersal shelf
Northern anchovy	<i>Engraulis mordax</i>	Epipelagic, coastal
California smoothtongue	<i>Leuroglossus stilbius</i>	Mesopelagic
Eared blacksmelt	<i>Lipolagus ochotensis</i>	Mesopelagic
Pacific hake	<i>Merluccius productus</i>	Semi-pelagic, shelf and slope
Pacific sardine	<i>Sardinops sagax</i>	Epipelagic
Croakers	Sciaenidae	Demersal, shelf
Rockfish	<i>Sebastes</i> spp.	Demersal and semi-pelagic, shelf and slope
Shortbelly rockfish	<i>Sebastes jordani</i>	Semi-demersal/pelagic, shelf and slope
Northern lampfish	<i>Stenobrachius leucopsarus</i>	Mesopelagic
Jack mackerel	<i>Trachurus symmetricus</i>	Epipelagic
Panama lightfish	<i>Vinciguerria lucetia</i>	Mesopelagic

even if detected, there will be limited ability to distinguish the effects of secular climate change or anthropogenic stressors from background climate variability.

A key impediment to the widespread inclusion of fishes in ocean observation programs is the perceived cost of fish surveys. Surveying a region's diverse fish populations is potentially a daunting prospect, given their typically patchy distributions over vast areas and varied benthic and pelagic habitats. Fishery surveys, whether carried out with nets, acoustics or visually are typically massive, costly endeavors.

However, ichthyoplankton surveys may afford a low-tech, relatively simple opportunity to quantitatively monitor much of the exceptional diversity of marine fishes. Most marine fishes broadcast their reproductive products into the open waters in great profusion (their fecundity is generally on the order of 10^4 – 10^7 offspring per female). The larvae of most species, even deep-sea fishes, generally inhabit the upper 200 m of the water column, so marine fishes inhabiting a diverse array of habitats from the

epipelagic to the deep sea can be quantitatively sampled with simple plankton nets during their egg and early larval stages.

Larval abundance is a function of spawning stock size, fecundity, and mortality during the early life history. Larval abundance alone is therefore not useful to estimate *absolute* stock biomass. However, numerous studies have found larval abundance to be significantly correlated with adult spawning stock biomass, particularly where stock size has varied significantly, thereby providing a useful *relative* index or proxy [12,17,19,28–30,38]. It should be noted that fish larvae are predominantly sampled with plankton nets for only a brief period after spawning during their pre-flexion stage, when motility and escapement are limited and they have experienced limited mortality. Because the taxonomy of fish during their larval stages is well-described for many parts of the world (e.g. [28,25]), quantitative larval fish time series can be developed for diverse assemblages of species.

Analyses of the CalCOFI ichthyoplankton time series, extending from 1951 to the present with quantitative data on > 400 taxa, indicate that such time series may provide considerable insight into the response of diverse regional fish communities to changing ocean conditions. Koslow et al. [19] reported a 63% decline in abundance across a broad suite of midwater fishes (24 taxa from 8 families) related to reduced midwater oxygen concentrations and expansion of the oxygen minimum zone. The CalCOFI data set also revealed an ~70% decline since about 1970 of a suite of taxa with cool-water affinities that included several of the most abundant taxa in the region [17,20]. These dramatic changes in major fish assemblages of the southern CCS were not apparent from commercial landings data or stock assessments carried out for commercial species [41].

Greater use of ichthyoplankton time series holds promise for vastly improving the ecological monitoring, assessment and management of the oceans [15]. However, ichthyoplankton time series are often perceived to be expensive and difficult to maintain, requiring large ship-based systematic sampling programs. In large part, this is because many such programs have as their primary objective to survey the egg and larval distribution of open-water broadcast spawners (e.g. sardine, anchovy, pollock, or mackerel)

Table 2

Taxa that contributed to PC 1 and 2 with their loadings on those PCs, equivalent to their correlation with the PC 1 time series. The habitat of the taxa loading highly on PC 1 is also shown. (From Koslow et al. [19,20]).

Taxon (PC 1)	Loading	Habitat	Taxon (PC 2)	Loading
<i>Diogenichthys atlanticus</i>	0.86	mesopelagic	<i>Ichthys lockingtoni</i>	0.75
<i>Protomyctophum crockeri</i>	0.85	mesopelagic	<i>Sebastes paucispinis</i>	0.71
<i>Ceratoscopelus townsendi</i>	0.83	mesopelagic	<i>Sebastes</i> spp.	0.70
<i>Cyclothone</i> spp.	0.83	mesopelagic	<i>Leuroglossus stilbius</i>	0.67
<i>Nannobranchium</i> spp.	0.81	mesopelagic	<i>Sebastes aurora</i>	0.63
<i>Sternoptychidae</i>	0.75	mesopelagic	<i>Peprilus simillimus</i>	0.61
<i>Stomias atriventer</i>	0.73	mesopelagic	<i>Engraulis mordax</i>	0.61
<i>Symbolophorus californiensis</i>	0.73	mesopelagic	<i>Merluccius productus</i>	0.56
<i>Chauliodus macouini</i>	0.71	mesopelagic	<i>Sciaenidae</i>	0.53
<i>Vinciguerria lucetia</i>	0.71	mesopelagic	<i>Pleuronichthys coenosus</i>	0.53
<i>Scopelarchidae</i>	0.70	mesopelagic	<i>Stenobrachius leucopsarus</i>	0.53
<i>Bathylagus wesethi</i>	0.69	mesopelagic	<i>Myctophidae</i>	0.52
<i>Bathylagus pacificus</i>	0.68	mesopelagic	<i>Melamphaes</i> spp.	0.51
<i>Scomber japonicus</i>	0.68	epipelagic	<i>Sebastolobus</i> spp.	0.50
<i>Paralepididae</i>	0.67	mesopelagic		
<i>Microstoma</i> spp.	0.67	mesopelagic	<i>Sardinops sagax</i>	−0.75
<i>Argentina sialis</i>	0.65	mesopelagic		
<i>Lipolagus ochotensis</i>	0.64	mesopelagic		
<i>Hygophum reinhardtii</i>	0.61	mesopelagic		
<i>Idiacanthus antrostomus</i>	0.57	mesopelagic		
<i>Gobiidae</i>	0.57	demersal nearshore		
<i>Myctophum nitidulum</i>	0.55	mesopelagic		
<i>Triphoturus mexicanus</i>	0.54	mesopelagic		
<i>Aristostomias scintillans</i>	0.53	mesopelagic		
<i>Myctophidae</i>	0.52	mesopelagic		
<i>Citharichthys</i> spp.	0.50	demersal shelf		
<i>Notolychnus valdiviae</i>	0.50	mesopelagic		

for stock assessment purposes [13,27,32,40]. The patchiness of ichthyoplankton distributions and variability of ichthyoplankton samples further contribute to the perception that extensive sampling is required to obtain reasonable confidence limits [34]. However, even extremely limited sampling programs, such as the Newport Hydrographic line, which consists of only two stations over the continental shelf off Oregon, have successfully delineated environmental relationships with larval fish abundance and diversity [5,7].

The purpose of this paper is to use the CalCOFI ichthyoplankton data set to examine how well a limited sampling program or reduced sample analysis captures trends in abundance of key species and multivariate patterns of community change. Such questions face many contemplating whether to start or continue ichthyoplankton sampling programs on more modest scales than CalCOFI or other large-scale ichthyoplankton survey programs. The effect of a reduced survey design—a single transect of varying length—is examined first, and the effect of a reduced time series—sampling for 10 or 20 years compared with the CalCOFI time series, which extends back to ~1950. How well random stratified sub-sampling of a broad-scale sampling program captures patterns of change in key species and the community is also examined. The option to analyze a portion of the samples from a large-scale program may need to be considered due to lack of funds to analyze the full data set or in order to more rapidly assess ecosystem status.

2. Materials and methods

CalCOFI ichthyoplankton data are available online: <http://coastwatch.pfeg.noaa.gov/erddap>, and the detailed CalCOFI ichthyoplankton survey protocols are provided in several sources [22,39,27]. Briefly, the core sampling area consists of stations along six transects that extend from San Diego to north of Point Conception and include coastal stations (~50 m depth), stations within the core of the California Current (CC), and stations ~250 to 550 km offshore (Fig. 1). At each station, ichthyoplankton samples were taken with a 505- μ m-mesh bongo-net towed obliquely from the surface to 10 m above the seafloor or up to ~210 m depth. The volume of water filtered was estimated using a flowmeter in the mouth of one of the nets. Fish eggs and larvae were removed, identified to the lowest taxon, and enumerated. Abundance (#/10 m²) was estimated for each taxon based on the counts, tow depth, and volume of water filtered.

Stations were sampled monthly to quarterly within the CalCOFI program. Seasonal means were determined and years not containing at least three seasons were removed from the analysis. Annual means derived from the seasonal means were used for analysis in order to minimize seasonal bias. Annual means were log-transformed to normalize the variance. Species were included in the analysis only if present for more than half of the survey years. There are gaps between 1966 and 1984 when only every third year was adequately sampled for inclusion in the time series. Principal component (PC) analyses carried out on this time series and sub-sets were based on the correlation matrix.

To examine the efficacy of sub-sampling, single transects, fractions of transects, or a stratified random selection of stations were selected. Subsamples of the CalCOFI data set were examined to assess how well reduced sampling captured patterns in both the annual mean abundance of individual taxa and multivariate (community-level) patterns in abundance. The standards for comparison were the annual means based on the 51 most consistently sampled stations along the 6 core transects and PC 1 and 2, based on the analysis of Koslow et al. [18–20]. For the first analyses, subsamples comprising single transects extending seaward from the coast out to station 40, 60, or to the end of each of the six core CalCOFI transects (as demarcated in Fig. 1) were selected. The number of stations in these sub-sampled transects ranged from 3 to 12, compared with the 51 stations along 6 transects in the original data set. This sub-sampling strategy was based on the assumption that a single transect extending out from the coast was a likely sampling design for observation programs with limited resources.

For a second set of analyses, stratified random subsamples were created by randomly selecting stations from 9 geographic strata based on a 3 alongshore $\times$ 3 onshore–offshore matrix: the two northern, two southern, and two central transects and the stations that were most inshore, that represented the core of the California Current, and that were most offshore. Three replicate analyses were carried out based on subsamples with 1, 2, or 3 stations per stratum.

To assess the performance of reduced sampling programs, the correlation between the sub-sampled and full CalCOFI ichthyoplankton time series was examined for each of the 12 most abundant species in the CalCOFI data set: sand dabs (*Citharichthys* spp.), northern anchovy (*Engraulis mordax*), California smooth-tongue (*Leuroglossus stilbius*), eared blacksmelt (*Lipolagus ochotensis*), Pacific hake (*Merluccius productus*), Pacific sardine (*Sardinops sagax*), croakers (*Sciaenidae*), rockfishes (*Sebastes* spp.), shortbelly rockfish (*Sebastes jordani*), northern lampfish (*Stenobrachius leucopsarus*), jack mackerel (*Trachurus symmetricus*), and Panama lightfish (*Vinciguerria lucetia*). These species are listed in Table 1 along with their habitats. They represent a diverse group

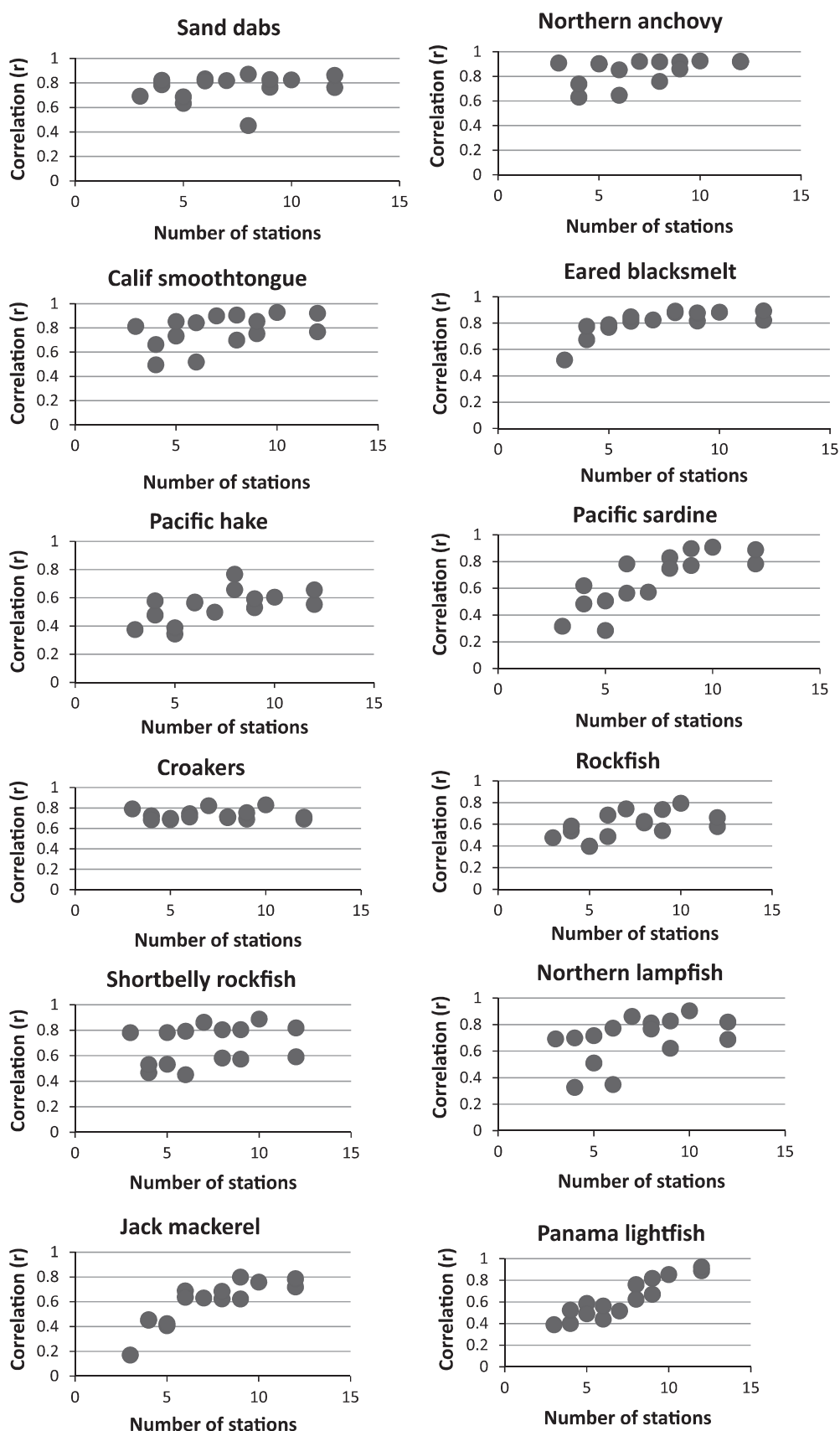


Fig. 2. The correlations (r) between time series (1951–2008) for the annual mean abundance of the 12 most abundant taxa in the CalCOFI ichthyoplankton data set based on the full data set and data sets derived from single transects extending to stations 40, 60, or the end of each of the 6 transects used in the present study (Fig. 1). This subsampling resulted in transects of 3–12 stations. The taxa are listed in Table 1 with their common and scientific names and habitat.

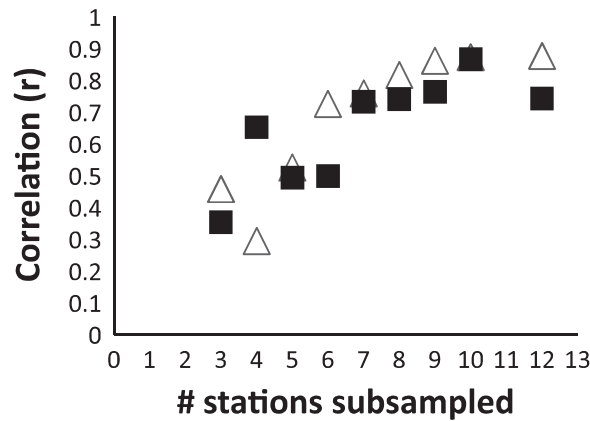


Fig. 3. The mean correlations (r) between PC time series (1951–2008) derived from the full CalCOFI ichthyoplankton data set and PCs derived individually from data sets with a given number of stations along a single transect. Individual data sets were based on the data obtained on each of the 6 transects from the stations that extended seaward to station 40, 60, or 80 (Fig. 1). Open triangles: correlations for PC 1; solid square: correlations for PC 2.

from coastal to oceanic habitats, including epipelagic, mesopelagic and demersal taxa, and taxa across several trophic levels and feeding guilds.

PCA analysis was also carried out on the sub-sampled time series based on the correlation matrix of log-transformed annual mean abundance for all taxa that were present in more than half the years from 1951 to 2011. PCA based on the correlation matrix normalizes the data of each taxon by its mean and standard deviation, minimizing bias toward more abundant and variable taxa. The first two PCs, which had eigenvalues > 1 , were extracted from each PCA and were correlated with PC 1–2 for the full ichthyoplankton data set published in Koslow et al. [19,20], which should be consulted for more details of the PCA (Table 2).

How time series length influenced correlations between sub-samples and the full CalCOFI data set was also examined. For these analyses, time series were based on the southernmost transect extending seaward of San Diego (line 93) (Fig. 1). These time series were then reduced to the last 10 or the last 20 years and compared with time series containing the full 49 years. Based on these reduced data sets, the influence of transect length and time series length was examined on correlations of abundance for the 12 most dominant taxa with the full data set and the correlations of the dominant patterns (PC 1–2) based on these reduced data sets with PC 1–2 derived from the full CalCOFI data set.

3. Results

3.1. Correlation with subsamples from single transects

The individual time series of abundance for the 12 most abundant species from the full data set and subsamples based on single transects of varying length (3–12 stations) were virtually all significantly correlated. The single exception was jack mackerel, for which the time series based on a transect of only 3 stations was not significantly correlated with the jack mackerel time series from the full data set. In general, the correlations between the full and reduced data sets declined most markedly with decreasing transect length for pelagic and mesopelagic species that spawn offshore, such as eared blacksmith, Pacific hake, Pacific sardine, jack mackerel and Panama lightfish (Fig. 2) [1,10,2,26]. Several of these taxa, such as jack mackerel, Pacific sardine, and Pacific hake are also noted for the interannual variability of their spawning distributions, so greater spatial coverage is required to adequately

sample their larval distributions. On the other hand, for species with predominantly nearshore distributions, such as sand dabs, northern anchovy, and croakers (Sciaenids), even just a few nearshore stations provided time series highly correlated with the full CalCOFI data set. For all taxa, time series based on a single transect of 10–12 stations were generally highly correlated with the full data set: correlations between 0.7 and 0.9.

PCA based on transect subsamples captured the dominant multivariate patterns (PC 1–2) of the full data set, with significant correlations between PCs derived from the full data set and from single transects even when transects were based on only 3 or 4 stations (Fig. 3). However, the correlations were generally fairly low (correlations (r) between 0.3 and 0.65 when the sub-sampled transects comprised 6 or fewer stations). Transects of 7 or more stations were generally required to obtain PC time series correlated with the original PC time series at levels of 0.7 and higher. It should be noted that the topography off southern California is characterized as a continental borderland, with depths > 200 m generally encountered close to the coast. In a region with a broad continental shelf, transects extending into deep waters would be required to observe patterns based on the mesopelagic fauna.

3.2. Correlation with stratified random sampling

How well time series derived from stratified random sub-sampling of the full data set represent the trends in abundance of dominant taxa and the primary multivariate patterns of community change was examined next.

As with sub-sampling based on single transects or fractions of a transect, the correlation between the annual mean abundance of the 12 most abundant species based on stratified random sub-samples and those in the full data set were significantly correlated. The CalCOFI survey area was divided into 9 strata, and for many taxa, the correlations between the full and sub-sampled data sets remained at roughly the same high level ($r=0.7$ – 0.9) regardless of whether 1, 2, or 3 samples per stratum were included (Fig. 4). The main exception to this was the croakers, which have a restricted nearshore distribution. In general, however, random-stratified sub-sampling was better able to capture the trends in abundance for dominant taxa than sub-sampling along a single, highly delimited transect. However, there was a minimum of 9 samples in the stratified random sub-samples but as few as 3 or 4 samples in the most restricted transect sub-samples.

The first two PCs derived from the full data set were highly positively correlated with their respective PCs derived from stratified random sub-samples ($r=0.8$ – 0.9) (Fig. 5). Increasing the number of samples per stratum between 1 and 3 had little effect on the correlation.

3.3. Reduced time series

When the time series was reduced to the last 10 or 20 years, the correlation for the 12 most abundant taxa between the full data set and data from a single transect varied by both species and length of the time series. For some taxa, there was a significant correlation between the full data set and sub-samples with both reduced time series (10 or 20 years) and a transect consisting of only 5 stations, particular among species with coastal spawning distributions: sand dabs and northern anchovy (Fig. 6). For taxa with more offshore spawning distributions, significant correlations were obtained when the transect extended across the CC (Pacific hake, Pacific sardine, and jack mackerel), while for some offshore species, such as Panama lightfish and northern lampfish, significant correlations with 10-year time series were only obtained when the transect extended beyond the core of the CC. As expected, with time series of 20 years, the time series based on the

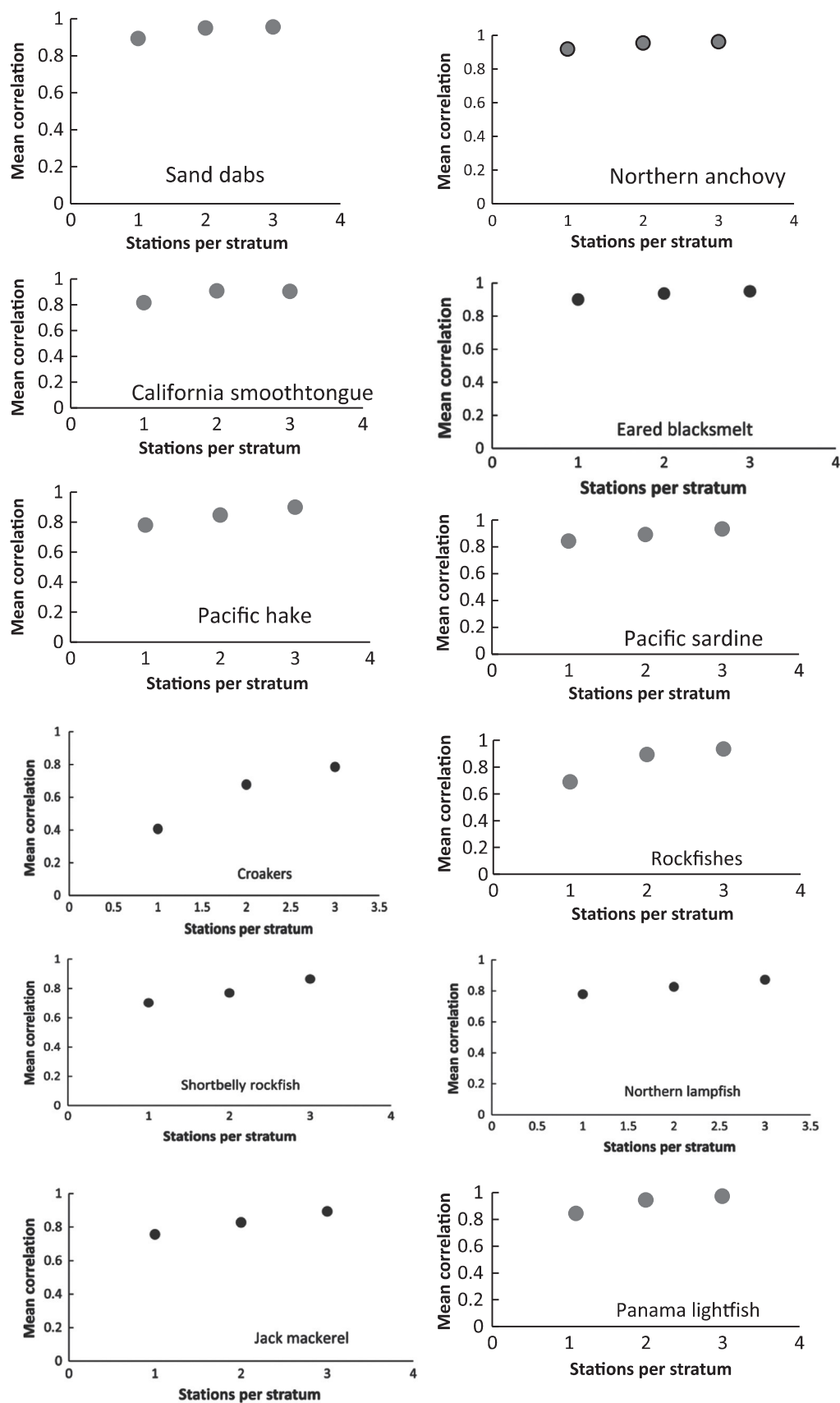


Fig. 4. The mean correlations (r) between time series (1951–2008) for the annual mean abundance of the 12 most abundant taxa in the CalCOFI ichthyoplankton data set based on the full data set and data sets derived from stratified random sub-samples based on 9 strata and 1, 2, or 3 samples per stratum.

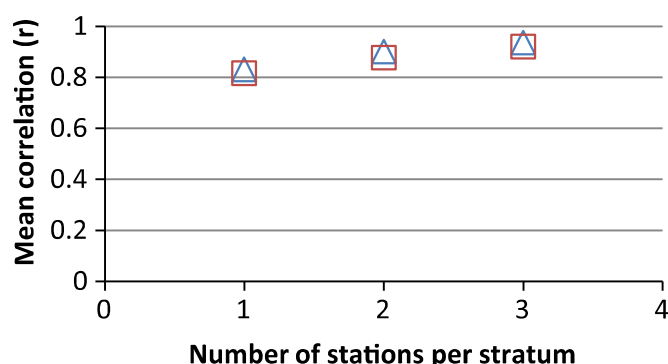


Fig. 5. The mean correlations (r) between PC time series (1951–2008) derived from the full CalCOFI ichthyoplankton data set and PCs derived individually from data sets derived from stratified random sub-samples based on 9 strata and 1, 2, or 3 samples per stratum. Open squares: PC 1; open triangles: PC 2.

full data set and a single transect were correlated for most taxa, regardless of transect length (Fig. 6). Exceptions included rockfishes and jack mackerel, whose time series were uncorrelated regardless of transect length, and Pacific sardine and shortbelly rockfish whose time series were only correlated when the transect extended across the CC.

There was no significant correlation between the first two PCs of the full data set and their corresponding PCs for subsamples with reduced time series that contained only the last 10 years of abundance data (Fig. 7). However, the first two PCs of the full data set were generally significantly correlated with their corresponding PCs based on sub-sampling along a single transect of varying length when the time series were based on the last 20 years of data, with the exception of a single sub-sample. Thus it generally seems possible to capture the multivariate patterns of the full data set with time series that are approximately 20 years or longer.

4. Discussion

Because most fishes inhabit the upper water column during their early life history and have limited avoidance capability, ichthyoplankton surveys provide a relatively low-cost, efficient means to monitor marine fish populations and communities. Only simple sampling gear (a plankton net with flowmeter) is required, and the taxonomy of marine fish larvae is well-developed for many parts of the world ocean. Larval abundance has been shown in many instances to serve as a good proxy for spawning stock biomass, even without the refinements of a full egg production survey [23,12,19,29–31]. As the CalCOFI program demonstrates, an ichthyoplankton sampling program that provides the requisite egg production data to carry out stock assessments for commercially-exploited species can also provide, with relatively little additional effort, time series for the abundance of virtually the entire region's fishes. Such surveys can also potentially provide information on changes in reproductive phenology [3] and spawning distribution.

Why then are there relatively few ichthyoplankton time series available for regional fish communities [15,16] in part, the problem may be institutional. Ichthyoplankton programs are maintained by fishery agencies, whose management mandate has typically focused on single-species approaches, despite frequent calls for ecosystem-based management [36]. The close association of ichthyoplankton programs with fisheries also led to the apparent association of ichthyoplankton time series with the extensive egg production survey designs that are required to assess stocks of wide-ranging broadcast spawners, such as anchovy, sardine, mackerel and so on. This combined with the well-known patchy distribution of fish eggs and larvae may have discouraged

institutions from establishing ocean observation programs that sample the ichthyoplankton.

It is demonstrated in this paper that more limited sampling programs (for example, a program based on sampling a relatively few stations along a single transect) can provide ichthyoplankton time series that capture the patterns of change observed in the dominant species (both fished and unfished) within the ecosystem, as well as multivariate patterns of change exhibited by assemblages of species. These findings are confirmed by the success of programs, such as the Newport Hydrographic line, which with only two shelf stations and 10 years of data has been able to detect significant change in fish assemblages of the northern CC [7]. This is possible because the signal:noise ratio for fishes in the CC system and elsewhere is quite high.

Dramatic fluctuations are commonly observed in commercial species [14,6,8]; however, it is less well-appreciated that the abundance of non-commercial fishes may also vary substantially on decadal time scales. As the CalCOFI ichthyoplankton time series has revealed, over the past half-century a suite of approximately 24 midwater taxa has declined ~63% from periods of high to low midwater oxygen concentrations, and many of the dominant taxa with cold-water affinities (both exploited and unexploited) have declined ~70% [18,19,21]. Thus, despite the patchy distribution of ichthyoplankton and the high variance associated with estimates of fish abundance based on egg and larval surveys, the changes in fish populations in the CC have been sufficiently large and sustained to emerge even from time series based on relatively few samples. Furthermore, by examining the abundance of assemblages of taxa rather than only a few taxa of commercial interest, each sample provides replicate observations of the state of the community. The dominant multivariate patterns of change (PCs 1–2) tend to be robust even with limited sampling.

Classic statistical treatments have typically framed the question of sample replication in terms of sample variance and the number of replicates required to establish adequate confidence limits for a single parameter. This is of paramount importance, for example, when determining the confidence limits for a stock assessment based on ichthyoplankton sampling. The issue of sample variance combined with the need to encompass a population's spawning distribution, which can prove extensive and highly variable from year to year, underlies the extensive sampling patterns associated with ichthyoplankton surveys for pelagic schooling species.

However, a different set of sampling issues confronts an ocean observation system. The key objectives of such programs are to detect changes in abundance of particular taxa or communities and to relate such change to changes in the ocean environment or anthropogenic stressors. Such programs are generally assumed to require at least 10, if not 20, years of data before much can be usefully said about interannual, much less decadal-scale, variability. Thus, multiple sets of years are often compared with each other (e.g. El Niño versus La Niña years, positive versus negative phase of an environmental index such as the PDO), which provide replication through the accumulation of years of data, not only replication based on stations within years. Furthermore, multivariate or community-based analysis examines the time series for a variety of taxa with similar biogeographic distributions or ecological requirements that may display coherent patterns of change. There is thus replication across taxa within each sample.

However, there is no free lunch in the design of ocean observation programs. Large elements of regional fish communities will not be observed if their habitats are not sampled. The continental borderland seafloor topography off southern California enables deepwater habitats and fish assemblages to be sampled within a short distance from the coast, so even very limited transects are able to sample diverse assemblages. However, in regions with broader shelf environments, a limited transect that

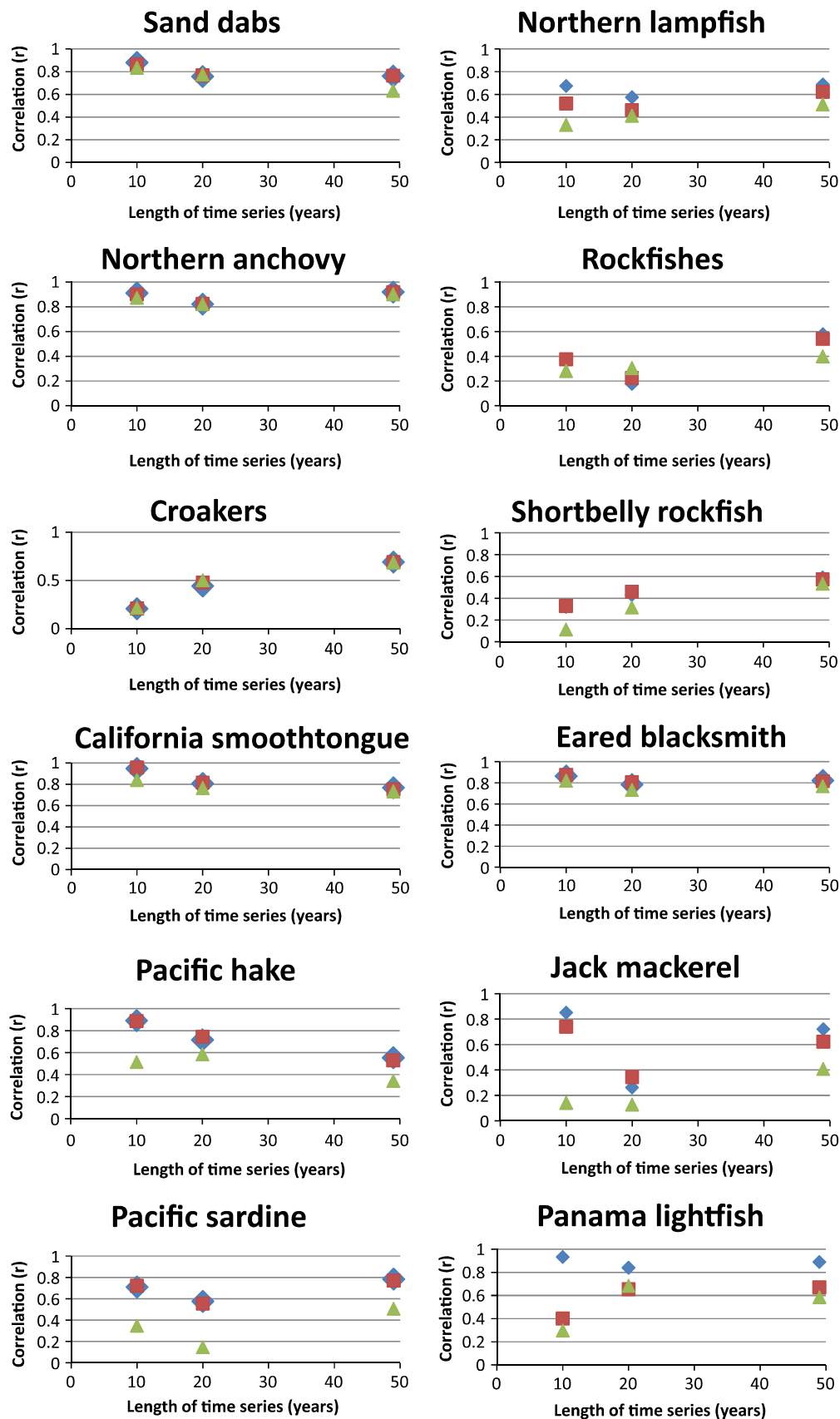


Fig. 6. Correlations between the log-transformed abundance of the 12 most abundant ichthyoplankton taxa in the CalCOFI data set from the full data set and from the southernmost transect (line 93) consisting of 5 stations (to station 40) (green triangles), 9 stations (to station 60) (red triangles) and the full transect of 12 stations (blue diamonds). The correlations were based on the last 10 and 20 years of the time series and the full time series. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

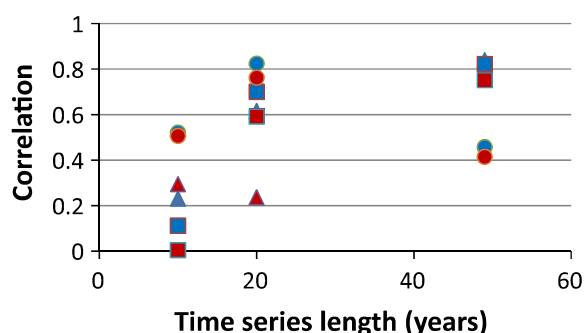


Fig. 7. The influence of time series length (years) and transect length (number of stations) on the correlation (r) of PC 1 (blue) and PC 2 (red) based on the full CalCOFI ichthyoplankton data set and data sets with transects comprised of 5 (circles), 9 (squares), and 12 (triangles) stations. The time series were comprised of the last 10 or 20 years of the time series or the full time series (49 data points). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

does not adequately sample the offshore environment will miss potentially important elements of regional fish communities [4]. It is worth noting that CalCOFI traditionally had few stations at depths < 50 m, and many nearshore fishes are not well represented in the time series [21].

Furthermore, these analyses demonstrate that increased sample size enables patterns to be recognized with shorter time series. Reducing the error bars around abundance estimates will obviously enhance confidence in any pattern that is observed.

In closing it is worth observing, first, that the use of ichthyoplankton time series can be transformative, pointing to the value of an ecosystem-based compared with a single-species approach to management. Analysis of the CalCOFI ichthyoplankton time series has revealed substantial changes in entire fish assemblages sharing common habitats (e.g. midwater fishes [19] or with common water mass affinities [12,18,21]).

Second, analysis of ichthyoplankton time series indicates that substantial change is already underway in the CC region. Such change will continue and almost certainly accelerate, along with the accelerating pace of climate change. It is incumbent upon us to observe the ecological implications of these changes. However, such change in the CC and elsewhere cannot be assessed without time series. Where no ichthyoplankton time series have yet been established, it is not too late to start; indeed, there is no better time than now. Where such programs are underway, it is critical that they are maintained. Where sample analysis has lagged, it would be prudent to continue the observations and to develop random stratified sub-sampling of the sample collections to extract the dominant patterns in the data. Ichthyoplankton time series provide a technologically simple, flexible and cost-effective tool that can be deployed in representative marine systems throughout the world's oceans as part of the fundamental underpinning for the science and management of marine fisheries and ecosystems in the face of a changing ocean environment.

5. Conclusions

There is a critical need to develop and maintain marine ecological time series in an era marked by climate change and other anthropogenic pressures. Fishes are highly diverse, highly valued, and play key roles in marine ecosystem; they are also sensitive indicators of natural and anthropogenic forcing. Ichthyoplankton surveys provide a low-tech means to quantitatively sample fish communities, with larval abundance generally a valuable proxy for adult spawning stock biomass, but egg and larval surveys for stock assessment are generally based on an extensive survey design and

costly. However, the present study, based on re-sampling the CalCOFI data set, indicates that sampling along a single transect across relevant habitats is sufficient to capture the dominant trends in populations of key species, as well as multispecies patterns of change. The study also indicates that for extensive surveys for which there are insufficient resources to analyze all samples, random stratified sub-sampling may also capture the dominant trends and multivariate patterns. These results indicate that ichthyoplankton surveys with varying designs could provide the basis for a low-cost, low-tech global system to monitor global fish communities and their response to changing ocean conditions.

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